

Particulate Matter Monitoring & Health Effects

Issue 2026-10-07 | Entry window 7 October 2026 (full day) | 14 records screened in scope

14

RECORDS IN SCOPE
(61 EXCLUDED)

7

OF 10 SUBTOPIC
BANDS POPULATED

2

EFFECT ESTIMATES
WITH A CI

4

TIER-A
RECORDS

5

RECORDS CARRIED ON
METADATA ALONE

Signal of the day

Congestion pricing cut $\text{PM}_{2.5}$ by $1.44 \mu\text{g m}^{-3}$ inside the zone — and the same day, a national analysis shows that averaging that kind of contrast away is how disparities disappear.

New York's Central Business District Tolling Program began in January 2025. Against 2022–2024, a difference-in-differences on NYCCAS real-time monitors attributes a $1.44 \mu\text{g m}^{-3}$ $\text{PM}_{2.5}$ reduction inside the congestion relief zone ($p < 0.001$; **1.33** allowing for spillover), while nearby non-zone sites rose a non-significant 0.50. Tropospheric NO_2 fell over 20 % across metro New York, but only **1.23** % of that is attributable to the programme ($p = 0.096$). Alongside it, block-level US emission intensities: medium- and heavy-duty trucks are under **11** % of vehicle kilometres but **46** % of NO_x and **50** % of $\text{PM}_{2.5}$ emission burden; people of colour face burdens **40** % above the national mean, and significant disparities appear in **85–89** % of counties, rural as much as urban. The finding that matters most is methodological: **spatial aggregation substantially attenuates the measured disparity. Why it matters for sensing:** both results are about resolution. A $1.4 \mu\text{g m}^{-3}$ step across a few square kilometres, and an equity gap that shrinks as you coarsen the grid, are the two clearest arguments yet in this series that network density is not a refinement — it decides what is visible.

What changed today

- **Black carbon, at lag0, across 212 million outpatient records.** 274 Chinese cities, 2013–2017: BC carried the strongest independent constituent associations, per IQR ($1.8 \mu\text{g m}^{-3}$) ranging from **+1.33** % (0.93–1.74) for chronic kidney disease to **+3.89** % (3.30–4.49) for COPD, and at lag0 rather than the lag0–1 typical of admissions. Outpatient visits are supply-sensitive, so a same-day signal may partly be care-seeking on visibly bad days.
- **Rural exposure misclassification, visible in the estimate.** 592 rural Californian children: an IQR (38 days) increase in prior-year days above $50 \mu\text{g m}^{-3}$ PM_{10} gave **+1.1 mmHg** DBP (0.8–1.5) and **+0.9 SBP** (0.3–1.5), with 5.8–5.9 percentage points higher risk of elevated SBP in the top short-term quartile. Exposure came from regulatory monitors that can be tens of kilometres away in a region whose sources are agricultural and local.
- **A network paper with no calibration.** A ZigBee mesh of low-cost CO and $\text{PM}_{2.5}$ nodes in Calabar reports mean $\text{PM}_{2.5}$ of $40 \mu\text{g m}^{-3}$, AQI 112.5 and a self-defined “performance index” of 76.5 % — with no co-location, no reference comparison and no humidity correction reported. West Africa's data gap is real; this does not close it.
- **A null worth having.** PRISm in 2,275 Kazakh adults: prevalence **12** % (10.7–13.3), higher in women, and *no* association with lifetime occupational vapour, gas, dust and fume exposure by job-exposure matrix — in a heavily industrial country. A European-derived JEM applied to Kazakh work histories is exactly the instrument that would null out a real effect.

Window continuity

The 6 October issue closed with `last_entry_date` at 6 October. This issue collects **7 October in full**, continuous with it. Unlike the 3–6 October issues, this day was **harvested and screened fresh on this run** — the cached backfill ended at 6 October. The 8 October daily and the W40 weekly (27 September–3 October) remain owed and follow.

Corpus analytics

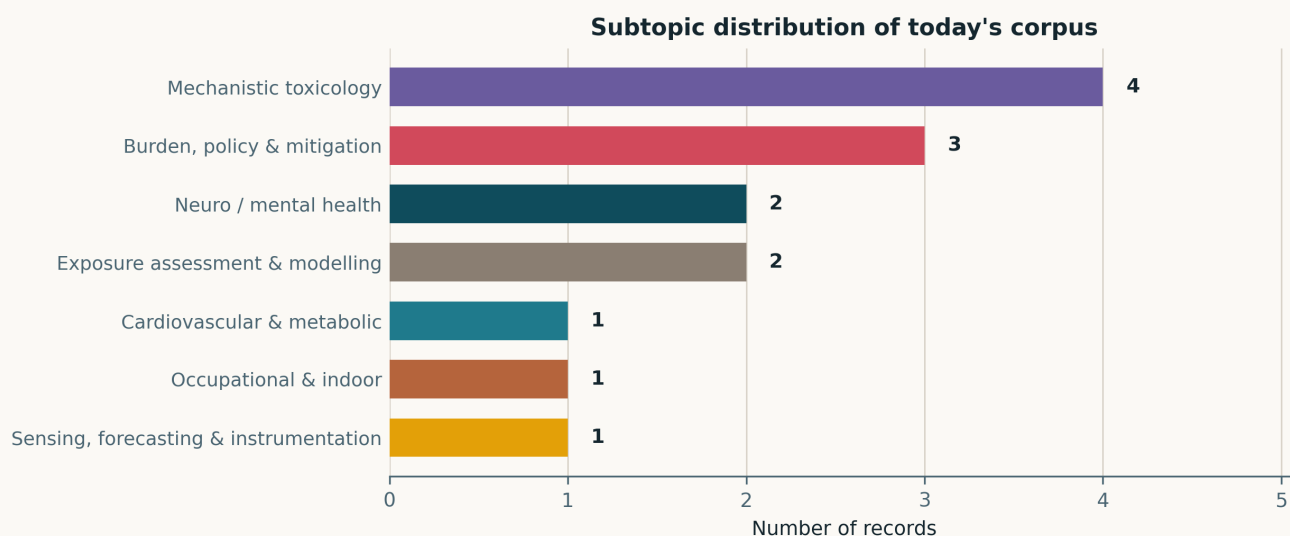


Fig. Subtopic assignment of the 14 in-scope records. **Seven of ten bands populated;** *Mechanistic toxicology* leads with **four** (three of them metadata-only or review), *Burden* three — and those three carry three of the day's four tier-A records — then two each for *Neuro* and *Exposure assessment*, and one each for *Cardiovascular*, *Occupational* and *Sensing*.

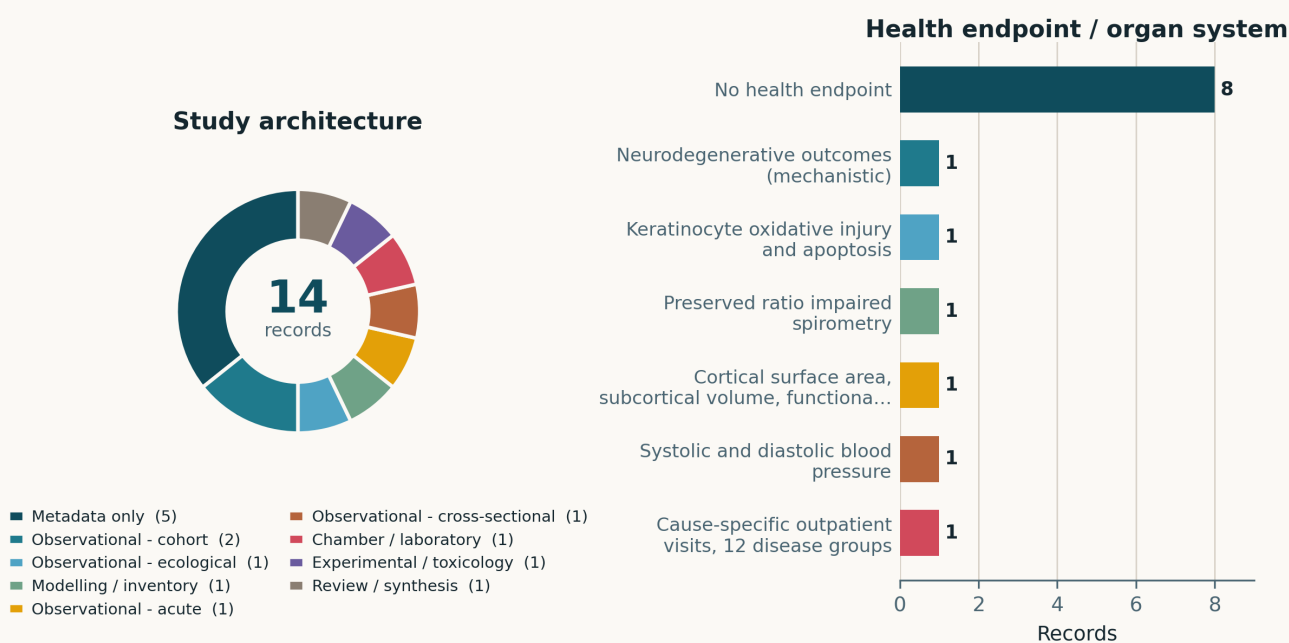


Fig. Left — architecture: **five** metadata-only deposits, two cohorts (the children's blood-pressure study and the ABCD imaging analysis), and one each for the ecological policy evaluation, the national inventory, the acute case-crossover, the cross-sectional spirometry survey, the chamber work, the toxicology experiment and the review. Right — the health endpoints are unusually spread: blood pressure, brain structure, outpatient visits, spirometry and the in-vitro endpoints each appear once.

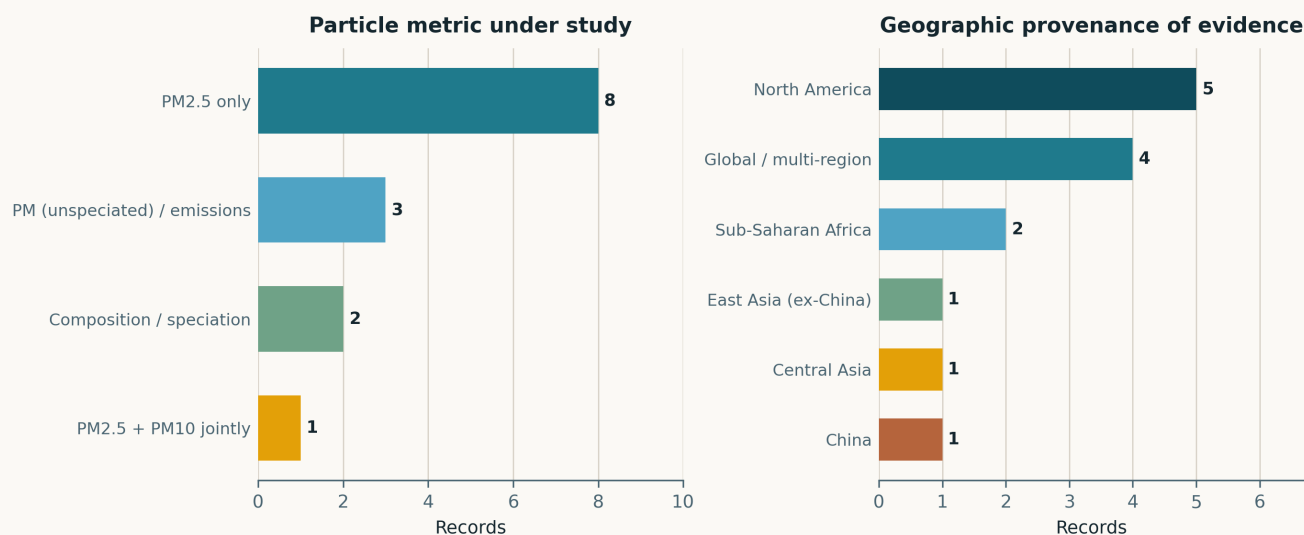


Fig. Left — particle metric: PM_{2.5} only **eight**, unspiciated PM or emissions three, composition/speciation two (the black-carbon constituent analysis and the manganese biomarker deposit) and PM_{2.5} + PM₁₀ jointly one. Right — geography: North America **five**, the most of any issue in this backfill, global / not stated four, Sub-Saharan Africa two (Calabar and the Ethiopian eruption), and one each for China, Central Asia and East Asia excluding China.

Life-course windows addressed today (records may span several stages)

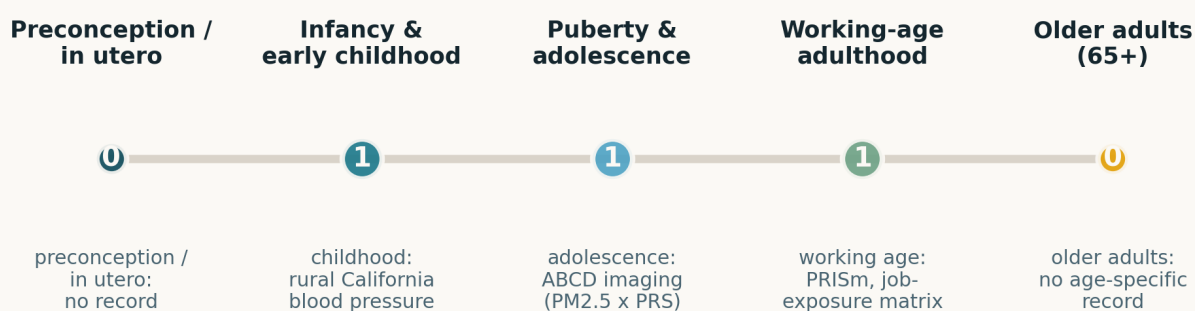


Fig. Life-course windows: childhood one (rural Californian blood pressure), adolescence one (the ABCD imaging cohort — the first adolescence-specific record since 26 September), working age one (the PRISm survey). In utero and older adults have no record.

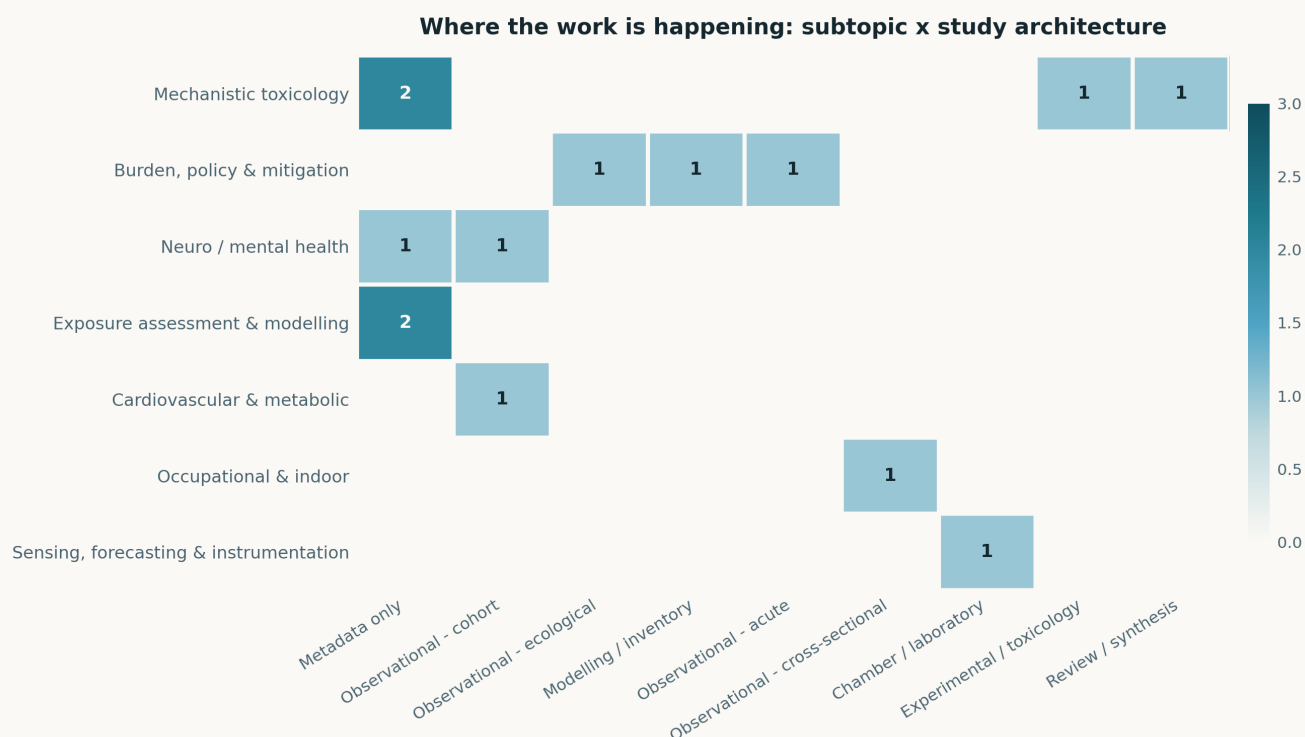


Fig. Subtopic × study architecture for the 14 records. The metadata-only block spans mechanistic toxicology, neuro and exposure assessment; burden holds three different architectures (ecological, inventory, acute) and no two records in it share a design — which is why the three agree only in direction.

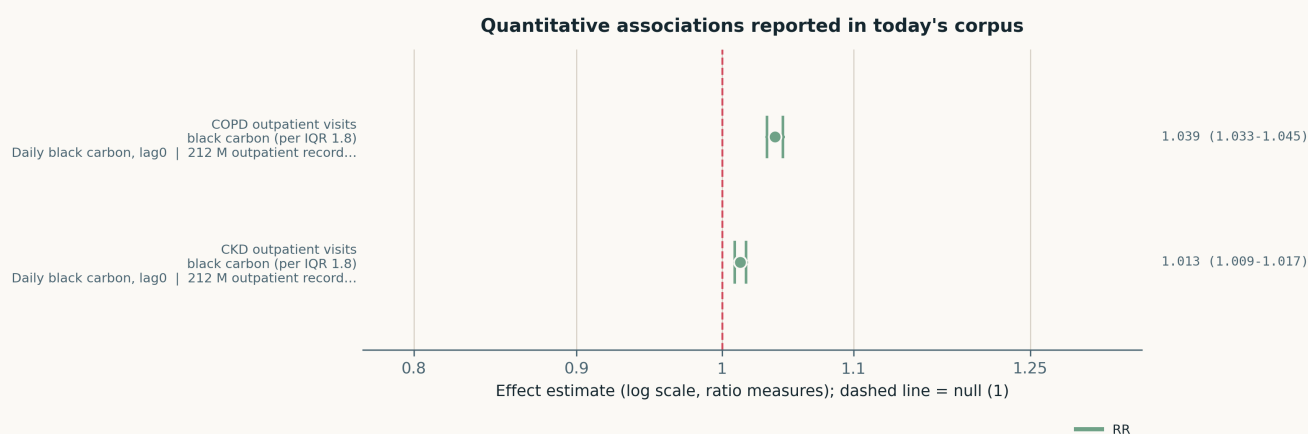


Fig. Two relative risks with 95 % CIs, log scale, both per interquartile range ($1.8 \mu\text{g m}^{-3}$) of daily black carbon at lag0, from the same 274-city case-crossover — COPD and chronic kidney disease, the extremes of the twelve disease groups reported. They share an exposure series and are not independent. The day's other quantitative results do not belong on a ratio scale: the New York policy effect is a concentration difference ($-1.44 \mu\text{g m}^{-3}$), the children's blood-pressure results are mean differences in mmHg and percentage-point risk differences, and the emission-burden disparities are ratios of emission intensity, not of risk.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Neuro / mental health

2 records

ABCD air pollution 2026 Schizophr Bull

The Relationship between Air Pollution Exposure, Genetic Liability for Schizophrenia, and Brain Outcomes in the ABCD Study

1,580 adolescents with genotype, age 9-10 exposure and structural plus resting-state MRI at 13-14; linear mixed-effects models with a schizophrenia polygenic score as moderator. Higher PM_{2.5} was associated with smaller right frontal pole surface area (p-FDR 0.039); the PRS interactions were non-monotonic - among low-PRS individuals, higher PM_{2.5} went with larger bilateral caudate volumes, while low-PRS/medium-PM_{2.5} showed smaller left caudate than medium/medium. Higher NO₂ was associated with stronger negative connectivity between networks. Limitation: a single surviving FDR-corrected main effect across a very large imaging search space, and interaction patterns that do not order monotonically in either PRS or exposure, is the signature of noise - the authors' own 'first test' framing is the right reading. Relevance: moderate - the exposure is a modelled annual surface at the residence of adolescents who move around a city all day.

doi: 10.1093/schbul/sbag138

PM_{2.5} and Mn biomarkers 2026 Environ Pollut

Differential associations of PM_{2.5} and particulate manganese with blood-based amyloid and tau biomarkers

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 7 Oct with no abstract. By title, PM_{2.5} mass and a specific metal constituent are contrasted against blood-based amyloid and tau - the preclinical Alzheimer biomarkers. Separating a constituent from total mass against a mechanistic biomarker is precisely the design the UFP dementia-mortality cohort (5 October) and the nitrate lung-cancer cohort (3 October) both point toward. Relevance: high; on the retry list.

doi: 10.1016/j.envpol.2026.129304

Cardiovascular & metabolic

1 record

Children's AIRE 2026 Int J Hyg Environ Health

Ambient particulate matter exposure and children's blood pressure in rural California, 2019-2022

592 children aged 6-8 at enrolment, blood pressure measured up to five times through spring 2022, linear mixed models. An IQR increase (38 days) in prior-year days above 50 $\mu\text{g m}^{-3}$ daily PM₁₀ was associated with +1.1 mmHg DBP (95% CI 0.8, 1.5) and +0.9 mmHg SBP (0.3, 1.5); the top quartile of short-term PM_{2.5} and PM₁₀ carried 5.8 (0.4, 11.2) and 5.9 (0.2, 11.6) percentage-point higher risk of elevated or hypertensive SBP against the lowest quartile. Limitation: exposure comes from regulatory monitors in a rural region where the nearest monitor can be tens of kilometres away and agricultural and wildfire sources are highly local, so non-differential misclassification is large and almost certainly biases toward the null; paediatric BP is noisy across visits. Relevance: high - this is the archetypal study whose exposure term a rural low-cost network would transform, and the 'days above threshold' metric suits an inexpensive, lower-accuracy node better than an annual mean does.

doi: 10.1016/j.ijheh.2026.114923

Mechanistic toxicology

4 records

Melatonin DPM2.5 2026 Ecotoxical Environ Saf*Melatonin protects keratinocytes from diesel-derived PM_{2.5}-induced damage by suppressing JNK/p38 MAPK signaling and caspase activation*

Diesel PM_{2.5} raised intracellular ROS, damaged macromolecules, disrupted mitochondrial membrane potential and drove apoptosis in HaCaT cells; melatonin scavenged radicals, restored viability and membrane potential, partly restored Bcl-2 against raised Bax and Bim, and suppressed caspase-9 and -3 and JNK/p38 MAPK activation, with a hairless-mouse arm. Limitation: the same template as the curcumin-nanosphere record on 3 October - a protective-agent paper in which the PM arm is a generic oxidant stimulus, with no composition, no dose-response against a deposited-dose estimate and no endotoxin control; diesel PM is a reference material, not an ambient aerosol. Relevance: low as mechanism; logged to keep the dermal-endpoint strand complete and because the DPM2.5 reference material is at least characterised, unlike most such papers.

doi: 10.1016/j.ecoenv.2026.120879

Env epigenetics review 2026 Front Epigenet Epigenom*Environmental epigenetic modifiers in neurodegenerative diseases: a systematic review of epigenomic mechanisms and therapeutic targets*

A PRISMA-structured review of DNA methylation, histone modification and non-coding RNA changes linking environmental neurotoxins - particulate matter among them - to neurodegeneration, with emphasis on developmental windows and life-course exposure. Limitation: PRISMA structure without quantitative synthesis, risk-of-bias assessment or any pooling, over a literature in which publication bias toward positive methylation findings is severe; particulate matter is treated interchangeably with metals and solvents despite incomparable exposure metrics. Relevance: low as evidence. Entered because the epigenetic mediator claim keeps appearing beside the dementia-mortality results in this backfill and it should be visible how thin the review base under it is.

doi: 10.3389/freae.2026.1833533

Mountain West wood smoke 2026 Part Fibre Toxicol*Household wood smoke exposure, lung black carbon deposition, and genomic instability in U.S. Mountain West residents*

METADATA ONLY. Metadata-only record: Crossref deposited title, authors and DOI on 7 Oct; neither PubMed, Europe PMC nor Semantic Scholar held an abstract, and Scholar Gateway is still refusing. By title this is an unusually direct chain - a household exposure, a measured internal dose (lung black carbon deposition) and a genomic endpoint in the same human subjects. Internal dose is what every ambient-monitoring epidemiology study substitutes a model for, so a paper that measures it is worth the retry. Relevance: high, the highest of the day's metadata-only records; flagged for full-text read.

doi: 10.1186/s12989-026-00702-8

Vocal fold fibroblasts 2026 Laryngoscope*Effects of Fine Particulate Matter (PM_{2.5}) in Cultured Human Vocal Fold Fibroblasts*

METADATA ONLY. Metadata-only record: Wiley deposited title, authors and DOI on 7 Oct with no abstract. By title, an in-vitro exposure of human vocal fold fibroblasts to PM_{2.5} - an unusual target tissue, and a plausible one given that the larynx sits in the deposition path of coarse and fine particles and that occupational voice disorders cluster in dusty trades. Relevance: low-moderate; on the retry list.

doi: 10.1002/lary.70976

Exposure assessment & modelling

2 records

Hayli Gubbi eruption 2026 Atmos Environ*Short-term Atmospheric Impacts of the Hayli Gubbi Volcanic Eruption 2025*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 7 Oct with no abstract. By title, the short-term atmospheric impact of the 2025 Hayli Gubbi eruption. Volcanic SO₂ and ash plumes are the natural experiment that tests whether a sparse monitoring network can detect and attribute a known, dated, large perturbation - and East African ground monitoring is close to absent, so the answer is informative either way. Relevance: moderate; on the retry list.

doi: 10.1016/j.atmosenv.2026.122406

N-containing OA 2026 Atmos Environ*Unveiling the Precursor-Specific Aqueous Formation Mechanisms of Nitrogen-Containing Organic Aerosols*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 7 Oct with no abstract. By title, a mechanistic account of how particular precursors form nitrogen-containing organic aerosol in the aqueous phase. Aqueous-phase secondary formation is humidity-dependent, which makes it the physical process underneath the humidity correction every optical low-cost sensor applies - the correction treats growth as hygroscopic swelling when some of it is mass that was not there before. Relevance: moderate-high; on the retry list.

doi: 10.1016/j.atmosenv.2026.122408

Sensing, forecasting & instrumentation

1 record

Calabar ZigBee WSN 2026 Research Square (preprint)*Design and Performance Evaluation of a ZigBee-Based Wireless Sensor Network for CO and PM_{2.5} Air Quality Monitoring in Calabar Metropolis*

A ZigBee mesh of low-cost CO and PM_{2.5} nodes, with analytical models for sensing, communication, energy use and dispersion exercised in MATLAB. Reported: reliable links, coverage growing with node count, a lifetime of about 4,065 transmissions, mean CO 30 ppm, mean PM_{2.5} 40 µg m⁻³, AQI 112.5, and a composite 'performance index' of 76.5%. Limitation: the central weakness is that no calibration, co-location or reference comparison is reported at all - a 40 µg m⁻³ mean from uncalibrated optical nodes in a humid tropical city is not a measurement, and a self-defined performance index is not a validation; the dispersion component is simulated, not observed. Relevance: moderate as a record of the deployment gap in West Africa, where ambient data is genuinely absent; low as evidence. This is the failure mode this watch exists to flag.

doi: 10.21203/rs.3.rs-10637757/v1

Occupational & indoor

1 record

Kazakhstan PRISm 2026 Front Public Health*Occupational associations of preserved ratio impaired spirometry (PRISm) in a population-based study*

2,275 adults from the general population with spirometry and a lifetime job-exposure matrix. PRISm prevalence was 12% (95% CI 10.7-13.3), higher in women (14%) than men (10%). Occupational airborne exposure including VGDF was not associated with PRISm, as ever-exposure or cumulative years; BMI was (adjusted OR 1.05 per unit). Limitation: a job-exposure matrix built for European job titles applied to Kazakh work histories misclassifies heavily and will null out a real association; cross-sectional design cannot separate PRISm from healthy-worker selection out of dusty jobs. Relevance: moderate - a clean negative in a heavily industrial country, and a direct argument for measured rather than matrix-assigned occupational exposure.

doi: 10.3389/fpubh.2026.1921126

Burden, policy & mitigation

3 records

NYC congestion pricing 2026 Environ Sci Technol Lett*Attributing PM_{2.5} and NO₂ Changes One Year after New York City's Congestion Pricing Policy*

The Central Business District Tolling Program began in January 2025; its first year is evaluated against 2022-2024 with difference-in-differences on NYCCAS real-time PM_{2.5} and on satellite NO₂. Within the congestion relief zone, PM_{2.5} fell by 1.44 $\mu\text{g m}^{-3}$ attributable to the policy ($p < 0.001$), or 1.33 $\mu\text{g m}^{-3}$ allowing for spillover, while nearby non-zone sites showed a non-significant 0.50 $\mu\text{g m}^{-3}$ rise. Tropospheric NO₂ fell over 20% across metropolitan New York against a 2018-2024 baseline, but only 1.23% was attributable to the programme ($p = 0.096$). Limitation: one treated zone and one post-period, so the parallel-trends assumption rests on three pre-years; the non-zone increase is the signal that displacement may be real but underpowered. Relevance: high - a 1.4 $\mu\text{g m}^{-3}$ step change across a few square kilometres is exactly the contrast a dense low-cost network resolves and a regulatory network, here a purpose-built survey, only barely does.

doi: 10.1021/acs.estlett.6c00610

US vehicle emission burden 2026 Environ Sci Technol*Spatially Refined Analysis of Emission Burdens and Equity from Light-, Medium-, and Heavy-Duty Vehicle Traffic in the United States*

Block-level on-road emission intensities for the whole United States. Medium- and heavy-duty trucks are under 11% of vehicle kilometres but 46% of NO_x and 50% of PM_{2.5} emission burden; truck PM_{2.5} is exhaust-dominated while tyre and brake wear are nearly half of light-duty PM_{2.5}. People of colour face burdens averaging 40% above the national mean (Asian populations over 60%), lower-income populations 20-25% higher, with significant disparities in 85-89% of counties, rural as well as urban. Critically, spatial aggregation attenuates the measured disparity. Limitation: this is an emission-intensity metric, not a concentration or an exposure - no dispersion, no background, no infiltration - so it cannot be read as an inhaled dose. Relevance: high - the aggregation result is the quantitative case for network density: resolution is not cosmetic, it determines whether a disparity is visible at all.

doi: 10.1021/acs.est.6c12227

BC outpatient risk 2026 Eco Environ Health*Black carbon as a major contributor to ambient PM_{2.5}-related cause-specific outpatient risks: implications for early warning and health benefits evaluation of air quality improvement*

Over 212 million daily outpatient records, 2013-2017, with conditional logistic regression per city pooled by random effects and three complementary models testing constituent independence. Black carbon carried the strongest independent associations, and at lag0 rather than the lag0-1 typical of admissions: per IQR (1.8 $\mu\text{g m}^{-3}$) increases ranged from 1.33% (0.93-1.74) for chronic kidney disease to 3.89% (3.30-4.49) for COPD, attributable fractions 2.38% to 6.12%. Both PM_{2.5}- and BC-attributable burdens fell substantially after the 2013 clean-air programme. Limitation: constituents come from a modelled reanalysis and are collinear; outpatient visits are supply-sensitive, and the lag0 finding could reflect care-seeking on visibly bad days rather than a faster biological response. Relevance: high - BC is measurable by low-cost aethalometry, and a same-day constituent signal is the one an operational early-warning network could actually act on.

doi: 10.1016/j.eehl.2026.100283

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
ABCD air pollution	The Relationship between Air Pollution Exposure, Genetic Liability for Schizophrenia, and Brain Outcomes in the ABCD Study	<i>Schizophr Bull</i>	Prospective cohort	Modelled annual PM _{2.5} , NO ₂ and O ₃ at age 9-10	USA (ABCD Study)
PM _{2.5} and Mn biomarkers	Differential associations of PM _{2.5} and particulate manganese with blood-based amyloid and tau biomarkers	<i>Environ Pollut</i>	Metadata only (no abstract)	Metadata only - PM _{2.5} and particulate manganese against blood amyloid and tau biomarkers (by title)	Not recoverable from metadata
Cardiovascular & metabolic					
Children's AIRE	Ambient particulate matter exposure and children's blood pressure in rural California, 2019-2022	<i>Int J Hyg Environ Health</i>	Prospective cohort	12-month and 7-day PM _{2.5} and PM ₁₀ from regulatory monitors linked to residence	USA (rural California)
Mechanistic toxicology					
Env epigenetics review	Environmental epigenetic modifiers in neurodegenerative diseases: a systematic review of epigenomic mechanisms and therapeutic targets	<i>Front Epigenet Epigenom</i>	Systematic review (PRISMA)	Particulate matter among heavy metals, gaseous pollutants and synthetic neurotoxicants (review)	Global
Melatonin DPM _{2.5}	Melatonin protects keratinocytes from diesel-derived PM _{2.5} -induced damage by suppressing JNK/p38 MAPK signaling and caspase activation	<i>Ecotoxicol Environ Saf</i>	Animal + in vitro	Diesel-derived PM _{2.5} applied to HaCaT keratinocytes and to HR-1 hairless mice	South Korea (laboratory)
Mountain West wood smoke	Household wood smoke exposure, lung black carbon deposition, and genomic instability in U.S. Mountain West residents	<i>Part Fibre Toxicol</i>	Metadata only (no abstract)	Metadata only - household wood smoke, lung black carbon deposition and genomic instability (by title)	USA (Mountain West)
Vocal fold fibroblasts	Effects of Fine Particulate Matter (PM _{2.5}) in Cultured Human Vocal Fold Fibroblasts	<i>Laryngoscope</i>	Metadata only (no abstract)	Metadata only - PM _{2.5} applied to cultured human vocal fold fibroblasts (by title)	Not recoverable from metadata
Occupational & indoor					
Kazakhstan PRISm	Occupational associations of preserved ratio impaired spirometry (PRISm) in a population-based study	<i>Front Public Health</i>	Observational - cross-sectional	Lifetime occupational vapour, gas, dust and fume exposure by job-exposure matrix; no PM measured	Kazakhstan (Aktobe, Karaganda, Kostanay)
Sensing, forecasting & instrumentation					
Calabar ZigBee WSN	Design and Performance Evaluation of a ZigBee-Based Wireless Sensor Network for CO and PM _{2.5} Air Quality Monitoring in Calabar Metropolis	<i>Research Square (preprint)</i>	Instrument development	Low-cost CO and PM _{2.5} nodes on an IEEE 802.15.4 ZigBee mesh, with MATLAB models of sensing, link, energy and dispersion	Nigeria (Calabar)
Exposure assessment & modelling					
Hayli Gubbi eruption	Short-term Atmospheric Impacts of the Hayli Gubbi Volcanic Eruption 2025	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - short-term atmospheric impacts of a 2025 eruption (by title)	Ethiopia (Hayli Gubbi, Afar)

First author	Focus	Journal	Design	Metric	Region
N-containing OA	Unveiling the Precursor-Specific Aqueous Formation Mechanisms of Nitrogen-Containing Organic Aerosols	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - precursor-specific aqueous formation of nitrogen-containing organic aerosol (by title)	Not recoverable from metadata
Burden, policy & mitigation					
BC outpatient risk	Black carbon as a major contributor to ambient PM _{2.5} -related cause-specific outpatient risks: implications for early warning and health benefits evaluation of air quality improvement	<i>Eco Environ Health</i>	Time-stratified case-crossover	Daily PM _{2.5} and constituents (black carbon, organic matter, sulfate, nitrate, ammonium), 274 cities	China (274 cities)
NYC congestion pricing	Attributing PM _{2.5} and NO ₂ Changes One Year after New York City's Congestion Pricing Policy	<i>Environ Sci Technol Lett</i>	Difference-in-differences evaluation	Real-time PM _{2.5} from the NYC Community Air Survey plus TROPOMI tropospheric NO ₂ columns	USA (New York City)
US vehicle emission burden	Spatially Refined Analysis of Emission Burdens and Equity from Light-, Medium-, and Heavy-Duty Vehicle Traffic in the United States	<i>Environ Sci Technol</i>	Modelling / inventory	Census-block NO _x and PM _{2.5} emission intensity from MOVES factors and street-level traffic, exhaust vs tyre/brake wear	USA (national, every census block)

Provenance and method

Entry window **7 October 2026**, single day, continuous with the 6 October issue, and the first day in this backfill harvested on the run that published it. Harvester legs run leg-by-leg: **PubMed** health 9, sensing **1**; **Europe PMC** 20 (11 after the AGRICOLA retro-index gate dropped 9); **Crossref** by **ISSN 43**; **arXiv** answered (148 kB) with no entry dated 7 October; **OpenAlex** HTTP 429 — the leg has now failed on every run since 26 September and should be considered dead rather than degraded. 64 raw deduplicated to **62**, all fresh. The **PubMed connector** ([EDAT] 7–8 October, both axes as separate queries) returned 23 PMIDs dated 7 October, of which 2 merged a PMID onto an existing DOI-only record and **13** were appended — **75** candidates screened, **14 admitted**, **61 rejected** with logged reasons (water and soil chemistry, microplastics, wastewater, measurement-science and remote-sensing engineering, and the usual broad-axis noise). **Five records carried on metadata alone** (36%), including the day's highest-relevance record (*Part. Fibre Toxicol.*, measured lung black-carbon deposition). The abstract-retry leg recovered **6 of 11** attempted. **Consensus** returned 10 calibration hits, **0 new**. **Scholar Gateway** fails with ACCESS_DENIED. DOI gate: **0 fail**, **0 warn**. Abstracts remain the copyright of their respective publishers.